Advanced Algebra

Section 1: Polynomial Theorems & Synthetic Division

1 If $(x-2)$ is a factor of $P(x) = 2x^3 - 3x^2 + kx - 10$, what is the value of k?

a. 1

b. 2

c. 3

d. 4

2 What is the quotient and remainder when $3x^3 + 4x^2 - x + 5$ is divided by $(x+2)$?

a. Quotient $3x^2 - 2x + 3$, remainder -1

b. Quotient $3x^2 + 2x - 3$, remainder 1

c. Quotient $3x^2 - 2x - 3$, remainder 11

d. Quotient $3x^2 + 10x + 19$, remainder 43

3 Find the sum of the cubes of the roots of $x^2 - 4x + 2 = 0$.

a. 32

b. 40

c. 48

d. 52

Section 2: Rational Equations & Partial Fractions

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4 Solve the rational equation: $\frac{x}{x-2} + \frac{2}{x+2} = \frac{8}{x^2-4}$.

a. $x = 2$

b. $x = -2$

c. $x = 2$ and $x = -2$

d. No real solution

5 Decompose $\frac{8x-4}{x^2-4}$ into partial fractions $\frac{A}{x-2} + \frac{B}{x+2}$. What is A?

a. 2

b. 3

c. 4

d. 5

6 Simplify the compound fraction: $\frac{\frac{1}{x+h} - \frac{1}{x}}{h}$ for $h \neq 0$.

a. $-\frac{1}{x(x+h)}$

b. $\frac{1}{x(x+h)}$

c. $-\frac{1}{x^2}$

d. $\frac{h}{x(x+h)}$

Section 3: Radical Equations & Fractional Exponents

7 Solve for x: $\sqrt{2x+9} - \sqrt{x+1} = 2$.

a. 0

b. 8

c. 0 and 8

d. No real solution

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8 If $x^{2/3} - 5x^{1/3} + 6 = 0$, what are all real values of x?

a. 8 and 27

b. 2 and 3

c. ± 8

d. ± 27

9 Simplify the radical quotient: $\frac{x-8}{\sqrt[3]{x}-2}$ for $x \neq 8$.

a. $x^{2/3} + 2x^{1/3} + 4$

b. $x^{2/3} - 2x^{1/3} + 4$

c. $x^{2/3} + 4$

d. $(\sqrt[3]{x} + 2)^2$

Section 4: Non-Linear Systems & Inequalities

10 Solve the non-linear system: $y = x^2 - 2x - 3$ and $y = 2x + 2$. What is the sum of the x-coordinates of the intersection points?

a. 2

b. 4

c. -4

d. 6

11 What is the solution set of the quadratic inequality $2x^2 - 5x - 3 > 0$?

a. $(-\infty, -\frac{1}{2}) \cup (3, \infty)$

b. $(-\frac{1}{2}, 3)$

c. $(-\infty, -3) \cup (\frac{1}{2}, \infty)$

d. $x > 3$

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12 For what values of m will the line $y = mx - 1$ NOT intersect the parabola $y = x^2$?

a. $(-2, 2)$

b. $[-2, 2]$

c. $(-\infty, -2) \cup (2, \infty)$

d. No such values of m

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